

Name _____

Worksheet 3-6: Solving Using Matrices

Learning Objectives:

- Use matrices to solve for unknown voltages and currents in a circuit

Materials Needed:

- Computer
- MATLAB – Either downloaded or MATLAB Online (<https://kr.mathworks.com/academia/tah-portal/university-of-utah-31156832.html>)

Part 1: System of Equations to Matrices

1. Convert the following system of equations to matrix format:

$$\frac{V_1 - 5V}{2k\Omega} + \frac{V_1 - V_2}{1k\Omega} + \frac{V_1}{500\Omega} - 2mA = 0$$

$$\frac{V_2 - V_1}{1k\Omega} + \frac{V_2}{4k\Omega} + \frac{V_2 - 2V}{1k\Omega} + 2mA = 0$$

$$\begin{bmatrix} & & & & \end{bmatrix} \begin{bmatrix} & & & & \end{bmatrix} = \begin{bmatrix} & & & & \end{bmatrix}$$

Part 2: Use MATLAB to Solve Matrices

There are many computer tools that you can use to solve matrices. MATLAB and Python are both designed to work primarily with matrices, so they are powerful programs for that. For this example, we will use MATLAB.

1. Use MATLAB to solve the following matrix equation:

$$\begin{matrix} \begin{bmatrix} 5 & -2 \\ 8 & 3 \end{bmatrix} & \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} & = & \begin{bmatrix} 1.5 \\ -6 \end{bmatrix} \\ A & B & & C \end{matrix}$$

- a) Define the matrices A, B, and C in MATLAB. Write the code you used to define the matrices below:

$A =$

$C =$

$B =$

- b) Enter the above matrices into MATLAB and report your answer for V_1 and V_2 :

$V_1 =$

$V_2 =$

2. Use MATLAB to solve the following matrix equation:

$$\begin{bmatrix} -4 & 2 & -13 \\ 1 & -1 & 1 \\ 1 & 2 & -4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -14 \\ 0 \\ -4 \end{bmatrix}$$

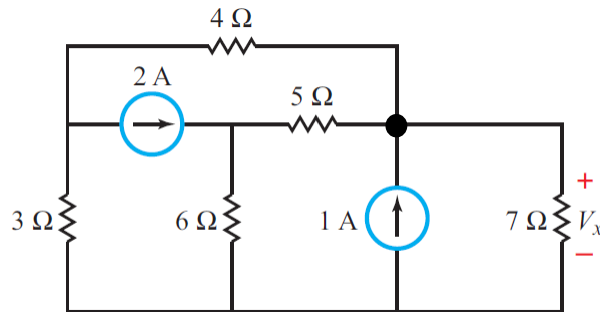
- a) Show what was typed in the computer: (screenshot or additional file attached to submission is fine)

- b) Then give the final numerical result:

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} =$$

Part 3: Solving Circuits with Matrices

1. (3.22 from textbook)
 - a) Create a matrix equation to represent the following circuit. Your matrix should include the variable V_x .

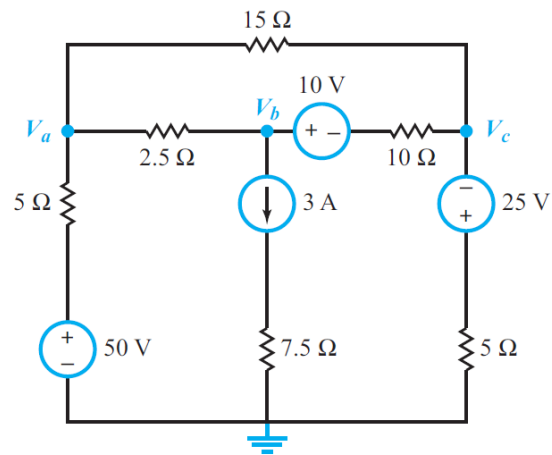


- b) Use a computer to solve the matrix and get the node voltages. (Attach/write your code for credit).

$$V_x = \underline{\hspace{2cm}}$$

2. (3.25 from textbook)

a) Create a matrix equation to represent the following circuit:



b) Use a computer to solve the matrix and get the node voltages. (Attach/write your code for credit).

$V_a =$ _____

$V_b =$ _____

$V_c =$ _____

Extra Credit: Designing and Solving a Circuit with MATLAB

1. Design a circuit with at least 3 unknown nodes. You can use as many voltage or current sources, and resistors as you would like.
 - a. Draw you circuit below (and label all nodes), setup the system of equations for the circuit, then solve for the unknown node voltages using MATLAB. Include a copy of your code for credit.